



Neet Test Hub Premium Test 12

Solution

Physics

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11 - A	12 - C	13 - A	14 - B	15 - C	16 - D	17 - D	18 - B	19 - C	20 - D
21 - D	22 - C	23 - B	24 - C	25 - C	26 - D	27 - A	28 - C	29 - C	30 - C
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41 - D	42 - C	43 - C	44 - A	45 - A					

Chemistry

46 - A	47 - B	48 - D	49 - D	50 - B	51 - A	52 - A	53 - D	54 - A	55 - A
56 - C	57 - D	58 - D	59 - C	60 - A	61 - C	62 - C	63 - B	64 - B	65 - D
66 - B	67 - C	68 - C	69 - A	70 - B	71 - B	72 - D	73 - A	74 - A	75 - B
76 - A	77 - D	78 - A	79 - D	80 - C	81 - D	82 - A	83 - C	84 - A	85 - C
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Biology - (Zoology)

91 - A	92 - D	93 - C	94 - A	95 - B	96 - C	97 - B	98 - C	99 - A	100 - D
101 - D	102 - B	103 - A	104 - D	105 - C	106 - A	107 - A	108 - A	109 - A	110 - B
111 - D	112 - B	113 - B	114 - A	115 - B	116 - D	117 - B	118 - D	119 - C	120 - A
121 - D	122 - D	123 - C	124 - C	125 - D	126 - B	127 - D	128 - D	129 - A	130 - D
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Biology - (Botany)

136 - A	137 - B	138 - A	139 - A	140 - B	141 - B	142 - B	143 - A	144 - C	145 - B
146 - D	147 - D	148 - D	149 - D	150 - A	151 - B	152 - A	153 - B	154 - A	155 - A
156 - B	157 - A	158 - C	159 - B	160 - A	161 - B	162 - B	163 - B	164 - A	165 - C
166 - D	167 - D	168 - B	169 - D	170 - C	171 - A	172 - B	173 - A	174 - B	175 - B
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Physics

1. AC alternating emf $E = 110\sqrt{2}\sin 100t$ volt is applied to a capacitor of $2\mu F$, the rms value of current in the circuit is.....mA.

A) 22 B) 20 C) 25 D) 30

Solution : (Correct Answer: A)

$$C = 2\mu F; \quad E = 110\sqrt{2}\sin(100t)$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 2 \times 10^{-6}}$$

$$= \frac{10000}{2} = 5000\Omega$$

$$i_0 = \frac{110\sqrt{2}}{5000}$$

$$i_{rms} = \frac{110\sqrt{2}}{5000\sqrt{2}}$$

$$= \frac{110}{5} \text{ mA}$$

$$= 22 \text{ mA}$$

2. In a series resonant LCR circuit, the voltage across R is 100 volts and $R = 1 \text{ k}\Omega$ with $C = 2 \mu f$. The resonant frequency ω is 200 rad/s. At resonance the voltage across L is

A) 40 v B) 250 v
C) $4 \times 10^{-3} \text{ v}$ D) $2.5 \times 10^{-2} \text{ v}$

Solution : (Correct Answer: B)

$$\text{Current } I = \frac{E}{Z}$$

$$\text{where } E = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\text{At resonance, } X_L = X_C$$

$$\therefore Z = R$$

$$\text{Again at resonance, } V_L = V_C$$

$$\therefore E = V_R$$

$$\therefore I = \frac{V_R}{R} = \frac{100}{1 \times 10^3} = 0.1 \text{ A}$$

$$\therefore I = \frac{V_R}{R} = \frac{100}{1 \times 10^3} = 0.1 \text{ A}$$

$$\therefore V_L = IL\omega = \frac{I}{C\omega} = \frac{0.1}{(2 \times 10^{-6}) \times (200)}$$

$$\therefore V_L = 250 \text{ volt.}$$

3. In an ac circuit, V and I are given by $V = 150 \sin(150t) \text{ V}$ and $I = 150 \sin(150t + \frac{\pi}{3}) \text{ A}$. The power dissipated in the circuit is.....W

A) 106 B) 150 C) 5625 D) 0

Solution : (Correct Answer: C)

$$\phi = \frac{\pi}{3}; E_{rms} = \frac{150}{\sqrt{2}}, I_{rms} = \frac{150}{\sqrt{2}}$$

$$P = E_{rms} I_{rms} \cos \phi$$

$$= \frac{150}{\sqrt{2}} \times \frac{150}{\sqrt{2}} \cos 60^\circ$$

$$= \frac{150 \times 150}{4} = 5625 \text{ W}$$

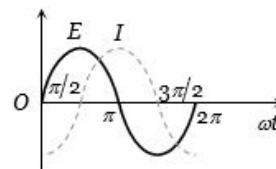
4. A 280 ohm electric bulb is connected to 200V electric line. The peak value of current in the bulb will be

A) About one ampere B) Zero
C) About two ampere D) About four ampere

Solution : (Correct Answer: A)

$$(a) i_{rms} = \frac{200}{280} = \frac{5}{7} \text{ A. So } i_0 = i_{rms} \times \sqrt{2} = \frac{5}{7} \times \sqrt{2} \approx 1 \text{ A.}$$

5. The variation of the instantaneous current (I) and the instantaneous emf (E) in a circuit is as shown in fig. Which of the following statements is correct



A) The voltage lags behind the current by $\pi/2$
B) The voltage leads the current by $\pi/2$
C) The voltage and the current are in phase
D) The voltage leads the current by π

Solution : (Correct Answer: B)

(b) At $t = 0$, phase of the voltage is zero, while phase of the current is $-\frac{\pi}{2}$ i.e., voltage leads by $\frac{\pi}{2}$

6. In a circuit, the value of the alternating current is measured by hot wire ammeter as 10 ampere. Its peak value will be.....A

A) 10 B) 20 C) 14.14 D) 7.07

Solution : (Correct Answer: C)

(c) Hot wire ammeter reads rms value of current.

$$\text{Hence its peak value} = i_{rms} \times \sqrt{2} = 14.14 \text{ amp}$$

7. If E_0 represents the peak value of the voltage in an ac circuit, the r.m.s. value of the voltage will be

A) $\frac{E_0}{\pi}$ B) $\frac{E_0}{2}$
C) $\frac{E_0}{\sqrt{\pi}}$ D) $\frac{E_0}{\sqrt{2}}$

Solution : (Correct Answer: D)

If E_0 represents the peak value of the voltage in an ac circuit then, we know,

$$E_{rms} = \frac{E_0}{\sqrt{2}}$$

8. The r.m.s. value of an ac of 50Hz is 10amp. The time taken by the alternating current in reaching from zero to maximum value and the peak value of current will be

A) $2 \times 10^{-2} \text{ sec}$ and 14.14 amp
B) $1 \times 10^{-2} \text{ sec}$ and 7.07 amp
C) $5 \times 10^{-3} \text{ sec}$ and 7.07 amp
D) $5 \times 10^{-3} \text{ sec}$ and 14.14 amp

Solution : (Correct Answer: D)

(d) Time taken by the current to reach the maximum value $t = \frac{T}{4} = \frac{1}{4\nu} = \frac{1}{4 \times 50} = 5 \times 10^{-3} \text{ sec}$

$$\text{and } i_0 = i_{rms} \sqrt{2} = 10\sqrt{2} = 14.14 \text{ amp}$$

9. The ratio of peak value and r.m.s value of an alternating current is

A) 1 B) $\frac{1}{2}$ C) $\sqrt{2}$ D) $1/\sqrt{2}$

Solution : (Correct Answer: C)

$$V_0 = V_{rms} \times \sqrt{2}$$

so,

$$\frac{V_0}{V_{ms}} = \sqrt{2}$$

10. The self inductance of a coil having 500 turns is 50 mH. The magnetic flux through the cross-sectional area of the coil while current through it is 8 mA is found to be

- A) $4 \times 10^{-4} \text{ Wb}$ B) 0.04 Wb
C) $4 \mu\text{Wb}$ D) 40 mWb

Solution : (Correct Answer: A)

(a)

The magnetic flux through an inductor is the self inductance of coil times the current through it.

Thus $\phi = LI$

$$= 50 \times 10^{-3} \text{ H} \times 8 \times 10^{-3} \text{ A}$$

$$= 4 \times 10^{-4} \text{ Wb}$$

11. A circular loop of radius 0.3 cm lies parallel to a much bigger circular loop of radius 20 cm. The centre of the smaller loop is on the axis of the bigger loop. The distance between their centres is 15 cm. If a current of 2.0 A flows through the smaller loop, then the flux linked with bigger loop $\times 10^{-11} \text{ weber}$

- A) 9.1 B) 6 C) 3.3 D) 6.6

Solution : (Correct Answer: A)

(a)

As we know, Magnetic flux, $\phi = BA$

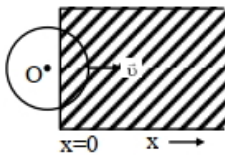
$$\frac{\mu_0(2)(20 \times 10^{-2})^2}{2[(0.2)^2 + (0.15)^2]} \times \pi (0.3 \times 10^{-2})^2$$

on solving

$$= 9.216 \times 10^{-11}$$

$$\approx 9.2 \times 10^{-11} \text{ weber}$$

12. A constant magnetic field of 1 T is applied in the $x > 0$ region. A metallic circular ring of radius 1 m is moving with a constant velocity of 1 m/s along the x -axis. At $t = 0$ s, the centre of O of the ring is at $x = -1$ m. What will be the value of the induced emf in the ring at $t = 1$ s? (Assume the velocity of the ring does not change.) (In V)



- A) 1 B) 4 C) 2 D) 0

Solution : (Correct Answer: C)

$$emf = BLV$$

$$= 1 \cdot (2R) \cdot 1$$

$$= 2V$$

13. Consider a circular coil of wire carrying constant current I , forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the circular coil area is given by ϕ_i . The magnetic flux through the area of the circular coil area is given by ϕ_0 . Which of the following option is correct?

- A) $\phi_i = -\phi_0$
B) $\phi_i = \phi_0$
C) $\phi_i < \phi_0$
D) $\phi_i > \phi_0$

Solution : (Correct Answer: A)

As current is flowing in the circular coil. The magnetic field in the region of the circular coil area is maximum than in the area of plane excluding the circular coil.

$$\Phi = \text{flux} = \vec{B} \cdot \vec{A}$$

$$\therefore \vec{B}_0 > \vec{B}_i$$

$$\Phi_0 = \vec{B}_0 \cdot \vec{A}$$

$$\Phi_i = \vec{B}_i \cdot \vec{A}$$

Magnetic field B in the circular region is more than outside $\therefore \vec{B}_0 > \vec{B}_i$

$$\therefore \Phi_0 > \Phi_i$$

14. The phase difference between the flux linked with a coil rotating in a uniform magnetic field and induced e.m.f. produce in it is

- A) π B) $\pi/2$ C) $\pi/3$ D) $-\pi/6$

Solution : (Correct Answer: B)

(b) $\phi = BA \cos \theta$ where θ is the angle between area and field

$$e = -\frac{d\phi}{dt} = -BA \sin \theta \cdot \frac{d\theta}{dt} = BA \left(\frac{d\theta}{dt} \right) \cos(90^\circ + \theta)$$

Here phase difference = 90° .

15. A circular disc of radius 0.2 meter is placed in a uniform magnetic field of induction $\frac{1}{\pi} \text{ Wb/m}^2$ in such a way that its axis makes an angle of 60° with $\vec{B}$. The magnetic flux linked with the disc is.....Wb

- A) 0.08 B) 0.01 C) 0.02 D) 0.06

Solution : (Correct Answer: C)

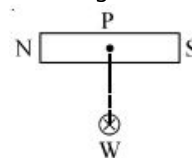
$$\vec{B} = \frac{1}{\pi} \left(\frac{\text{Wb}}{\text{m}^2} \right)$$

Area of the disc normal to B is $\pi R^2 \cos 60^\circ$

Flux = $B \times$ Area normal

$$\therefore \text{Flux} = \frac{1}{2} \times 0.04 = 0.02 \text{ Wb}$$

16. Figure shows a bar magnet and a long straight wire W , carrying current into the plane of paper. Point P is the point of intersection of axis of magnet and the line of shortest distance between magnet and the wire. If P is the midpoint of the magnet, then which of the following statements is correct?



- A) magnet experiences a torque in clockwise direction
B) magnet experiences a torque in anticlockwise direction
C) magnet experiences a force, normal to the line of shortest distance
D) magnet experiences a force along the line of shortest distance

Solution : (Correct Answer: D)

There is a long straight wire W , the current flowing direction in the wire is into the plane of paper and the point of intersection on the axis of magnet is P . If the midpoint of bar magnet is P then the distance

from the point P to the wire will be the shortest distance.

So, bar magnet will experience a force along the line of shortest distance.

17. A circular loop of radius R carrying current I lies in $x - y$ plane with its centre at origin. The total magnetic flux through $x - y$ plane is

A) Directly proportional to I
 B) Directly proportional to R
 C) Directly proportional to R^2
 D) Zero

Solution : (Correct Answer: D)

(d) Circular loop behaves as a magnetic dipole whose one surface will be N -pole and another will be S -pole. Therefore magnetic lines of force emerge from N will meet at S . Hence total magnetic flux through $x - y$ plane is zero.

18. The unit of magnetic flux is

A) Weber/m B) Weber C) Henry D) Ampere/m

Solution : (Correct Answer: B)

(b) Magnetic flux is magnetic field per unit area that is tesla per square metre which is defined as weber.

19. Due to a small magnet intensity at a distance x in the end on position is 9 Gauss . What will be the intensity at a distance $\frac{x}{2}$ on broad side on position..... Gauss

A) 9 B) 4
 C) 36 D) 4.5

Solution : (Correct Answer: C)

(c) In $C.G.S.$ $B_{axial} = 9 = \frac{2M}{x^3} \dots (i)$

$$B_{equatorial} = \frac{M}{(\frac{x}{2})^3} = \frac{8M}{x^3} \dots (ii)$$

From equation (i) and (ii) $B_{equatorial} = 36 \text{ Gauss}$.

20. Dip angle in vertical plane at an angle $\cos^{-1} \left(\frac{1}{\sqrt{2}} \right)$ from magnetic meridian is 60° , then actual dip at that place

A) $\tan^{-1} \left(\frac{\sqrt{3}}{2} \right)$
 B) $\tan^{-1} \left(\frac{1}{\sqrt{6}} \right)$
 C) $\tan^{-1} (1)$
 D) $\tan^{-1} \left(\sqrt{\frac{3}{2}} \right)$

Solution : (Correct Answer: D)

$$\alpha = \cos^{-1} \frac{1}{\sqrt{2}}, \Rightarrow \cos \alpha = \frac{1}{\sqrt{2}}, \theta' = 60^\circ$$

$$\tan \theta' = \tan \theta / \cos \alpha$$

$$\text{so } \tan \theta = \tan \theta' \cos \alpha$$

$$= (\tan 60^\circ) \frac{1}{\sqrt{2}} = \sqrt{3} \times \frac{1}{\sqrt{2}}$$

$$\theta = \tan^{-1} \left(\sqrt{\frac{3}{2}} \right)$$

21. Which statement is correct

A) Magnetic field of current element is perpendicular to position vector
 B) Electric field of point charge is along position vector
 C) Magnetic mono pole physically does not exist
 D) All of above

Solution : (Correct Answer: D)

22. Due to a small magnet, intensity at a distance x in the end on position is 9 gauss . What will be the intensity at a distance $\frac{x}{2}$ on broad side on position..... gauss

A) 9 B) 4 C) 36 D) 4.5

Solution : (Correct Answer: C)

$$\text{In } C.G.S. B_{axial} = 9 = \frac{2M}{x^3} \dots (i)$$

$$B_{equatorial} = \frac{M}{(\frac{x}{2})^3} = \frac{8M}{x^3} \dots (ii)$$

From equation (i) and (ii), $B_{equatorial} = 36 \text{ gauss}$.

23. A ferromagnetic material is heated above its curie temperature. Which one is a correct statement

A) Ferromagnetic domains are perfectly arranged
 B) Ferromagnetic domains become random
 C) Ferromagnetic domains are not influenced
 D) Ferromagnetic material changes itself into diamagnetic material

Solution : (Correct Answer: B)

(b) On heating, different domains have net magnetisation in them which are randomly distributed. Thus the net magnetisation of the substance due to various domains decreases to minimum.

24. Tangent galvanometer is used to measure

A) Charge B) Angle
 C) Current D) Magnetic intensity

Solution : (Correct Answer: C)

(c) When a current is passed through the galvanometer coil, then a magnetic field B is produced at right angles to the plane of the coil, i.e., at right angles to the horizontal component of earth's magnetic field H . Under the influence of two crossed magnetic fields B and H , the magnetic needle of galvanometer undergoes a deflection θ which is given by the tangent law. Using tangent law, we can find a relation

$$I \propto \tan \theta$$

Which clearly indicates that tangent galvanometer is an instrument used for detection of electric current in a circuit.

25. A thin rectangular magnet suspended freely has a period of oscillation equal to T . Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its period of oscillation is T' , then ratio $\frac{T'}{T}$ is

A) $\frac{1}{4}$ B) $\frac{1}{2\sqrt{2}}$ C) $\frac{1}{2}$ D) 2

Solution : (Correct Answer: C)

$$\text{Oscillation of } n^{\text{th}} \text{ part of magnet } T' = \frac{T}{n}$$

$$\Rightarrow \frac{T'}{T} = \frac{1}{n}; \text{ here } n = 2 \text{ so } \frac{T'}{T} = \frac{1}{2}.$$

26. Two lines of force due to a bar magnet

A) Intersect at the neutral point
 B) Intersect near the poles of the magnet
 C) Intersect on the equatorial axis of the magnet
 D) Do not intersect at all

Solution : (Correct Answer: D)

(d)

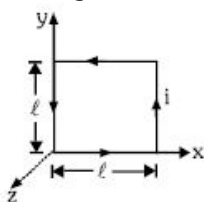
27. Magnetic lines of force due to a bar magnet do not intersect because

A) A point always has a single net magnetic field
 B) The lines have similar charges and so repel each other
 C) The lines always diverge from a single point
 D) The lines need magnetic lenses to be made to intersect

Solution : (Correct Answer: A)

Magnetic lines of force due to a bar magnet do not intersect because a point always has a single net magnetic field.

28. The magnetic field existing in a region is given by $\vec{B} = B_0 \left(5 + \frac{x}{\ell}\right) \hat{k}$. A square loop of edge ℓ and carrying a current i is placed with its edges parallel to $x - y$ axes. Find the magnitude of the net magnetic force experienced by the loop



A) $5 B_0 i \ell$ B) $6 B_0 i \ell$
 C) $B_0 i \ell$ D) Zero

Solution : (Correct Answer: C)

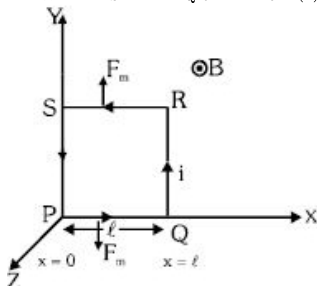
$$\vec{F}_{SR} = -\vec{F}_{PQ}$$

$$\vec{F}_{SR} + \vec{F}_{PQ} = 0$$

$$\vec{F}_{SP} = B_0 \left(5 + \frac{0}{\ell}\right) I \ell = 5 B_0 I \ell (-\hat{i})$$

$$\vec{F}_{QR} = B_0 \left(5 + \frac{\ell}{\ell}\right) I \ell = 6 B_0 I \ell (+\hat{i})$$

$$\vec{F}_{\text{net}} = \vec{F}_{SP} + \vec{F}_{QR} = 6 B_0 I \ell (\hat{i}) + 5 B_0 I \ell (-\hat{i}) = B_0 I \ell (\hat{i})$$



29. A charge particle of charge q and mass m is accelerated through a potential diff. V volts. It enters a region of orthogonal magnetic field B . Then radius of its circular path will be

A) $\sqrt{\frac{Vm}{2qB^2}}$
 B) $\frac{2Vm}{qB^2}$
 C) $\sqrt{\frac{2Vm}{q}} \left(\frac{1}{B}\right)$
 D) $\sqrt{\frac{Vm}{q}} \left(\frac{1}{B}\right)$

Solution : (Correct Answer: C)

$$R = \frac{mv}{qB} = \frac{\sqrt{2m(KE)}}{qB} = \frac{\sqrt{2m(qV)}}{qB}$$

30. The pole strength of a bar magnet is $48 \text{ ampere} - \text{metre}$ and the distance between its poles is 25 cm . The moment of the couple by which it can be placed at an angle of 30° with the uniform magnetic intensity of flux density $0.15 \text{ Newton/ampere} - \text{metre}$ will be..... $\text{Newton} \times \text{metre}$

A) 12 B) 18
 C) 0.9 D) None of the above

Solution : (Correct Answer: C)

$$(c) \tau = MB \sin \theta = 48 \times 25 \times 10^{-2} \times 0.15 \times \frac{1}{2} = 0.9 \text{ N} \times \text{m}$$

31. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron

A) will turn towards right of direction of motion
 B) will turn towards left of direction of motion
 C) speed will increase
 D) speed will decrease

Solution : (Correct Answer: D)

Force on electron due to electric field,

$$\vec{F}_E = -e\vec{E}$$

Force on electron due to magnetic field,

$$\vec{F}_B = -e(\vec{v} \times \vec{B}) = 0$$

since $\vec{v}$ and $\vec{B}$ are in the same direction.

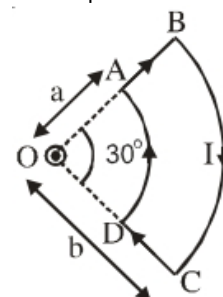
Total force on the electron,

$$\vec{F} = \vec{F}_E + \vec{F}_B = -e\vec{E}$$

Electric field opposes the motion of the electron, hence speed of the electron will decrease.

32. A current loop $ABCD$ is held fixed on the plane of the paper as shown in the figure. The arcs BC (radius = b) and DA (radius = a) of the loop are joined by two straight wires AB and CD . A steady current I is flowing in the loop. Angle made by AB and CD at the origin O is 30° . Another straight thin wire with steady current I_1 flowing out of the plane of the paper is kept at the origin.

The magnitude of the magnetic field (B) due to the loop $ABCD$ at the origin (O) is :



A) 0
 B) $\frac{\mu_0 I (b-a)}{24ab}$
 C) $\frac{\mu_0 I}{4\pi} \left[\frac{b-a}{ab} \right]$
 D) $\frac{\mu_0 I}{4\pi} \left[2(b-a) + \frac{\pi(a+b)}{3} \right]$

Solution : (Correct Answer: B)

The magnetic field at O due to current in DA is

$$B_1 = \frac{\mu_0}{4\pi} \frac{I}{a} \times \frac{\pi}{6} \quad (\text{directed vertically upwards})$$

The magnetic field at O due to current in BC is

$$B_2 = \frac{\mu_0 I}{4\pi b} \times \frac{\pi}{6} \quad (\text{directed vertically downwards})$$

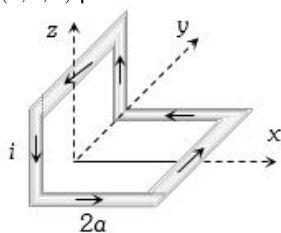
The magnetic field due to current AB and CD at O is zero.

Therefore the net magnetic field is

$$B = B_1 - B_2 \quad (\text{directed vertically upwards})$$

$$= \frac{\mu_0 I}{4\pi a} \frac{\pi}{6} - \frac{\mu_0 I}{4\pi b} \times \frac{\pi}{6} = \frac{\mu_0 I}{24} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{\mu_0 I}{24ab} (b - a)$$

33. A non-planar loop of conducting wire carrying a current I is placed as shown in the figure. Each of the straight sections of the loop is of length $2a$. The magnetic field due to this loop at the point $P(a, 0, a)$ points in the direction

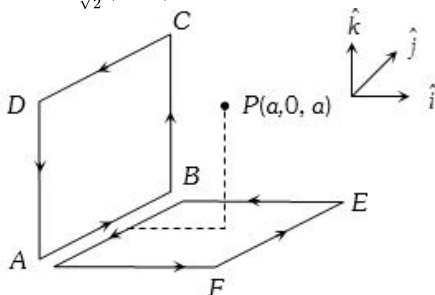


- A) $\frac{1}{\sqrt{2}}(-\hat{j} + \hat{k})$
 B) $\frac{1}{\sqrt{3}}(-\hat{j} + \hat{k} + \hat{i})$
 C) $\frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$
 D) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$

Solution : (Correct Answer: D)

(d) The magnetic field at $P(a, 0, a)$ due to the loop is equal to the vector sum of the magnetic fields produced by loops $ABCD$ and $AFEBA$ as shown in the figure.

Magnetic field due to loop $ABCD$ will be along $\hat{i}$ and due to loop $AFEBA$, along $\hat{k}$. Magnitude of magnetic field due to both the loops will be equal. Therefore, direction of resultant magnetic field at P will be $\frac{1}{\sqrt{2}}(\hat{i} + \hat{k})$.



34. A current of I ampere is passed through a straight wire of length 2.0 metres. The magnetic field at a point in air at a distance of 3 metres from either end of wire and lying on the axis of wire will be

- A) $\frac{\mu_0}{2\pi}$ B) $\frac{\mu_0}{4\pi}$
 C) $\frac{\mu_0}{8\pi}$ D) Zero

Solution : (Correct Answer: D)

(d) The magnetic field at any point on the axis of wire be zero.

35. When a certain length of wire is turned into one circular loop, the magnetic induction at the centre of coil due to some current flowing is B_1 . If the same wire is turned into three loops to make a circular coil, the magnetic induction at the center of this coil for the same current will be

- A) B_1 B) $9B_1$

- C) $3B_1$ D) $27B_1$

Solution : (Correct Answer: B)

$$(b) B' = n^2 B = (3)^2 B = 9B$$

36. A millimeter of range 10 mA has a coil of resistance 1Ω . To use it as voltmeter of range 10 volt, the resistance that must be connected in series with it, will be Ω

- A) 999 B) 99 C) 1000 D) None of these

Solution : (Correct Answer: A)

$$(a) R = \frac{V}{I_g} - G = \frac{10}{10 \times 10^{-3}} - 1 = 999 \Omega.$$

37. When a conducting soap bubble is negatively charged then

- A) Its size starts varying arbitrarily
 B) It expands
 C) It contracts
 D) No change in its size takes place

Solution : (Correct Answer: B)

(b)

Due to repulsion force between diametrically opposite wall, it expands.

38. A and B are two points on the axis and the perpendicular bisector respectively of an electric dipole. A and B are far away from the dipole and at equal distance from it. The field at A and B are $\vec{E}_A$ and $\vec{E}_B$.

- A) $\vec{E}_A = \vec{E}_B$
 B) $\vec{E}_A = 2\vec{E}_B$
 C) $\vec{E}_A = -2\vec{E}_B$
 D) $|E_B| = \frac{1}{2}|E_A|$, and $\vec{E}_B$ is perpendicular to $\vec{E}_A$

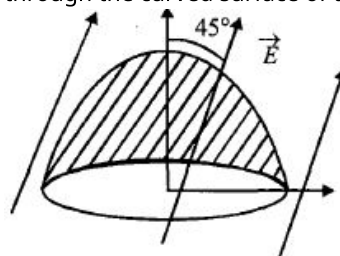
Solution : (Correct Answer: C)

Electric field due to dipole at axial position is

$$E_{\text{axial}} = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3} \quad (\text{along the dipole moment})$$

$$\text{Hence, } \vec{E}_A = -2\vec{E}_B$$

39. The flat base of a hemisphere of radius a with no charge inside it lies in a horizontal plane. A uniform electric field $\vec{E}$ is applied at an angle $\frac{\pi}{4}$ with the vertical direction. The electric flux through the curved surface of the hemisphere is



- A) $\pi a^2 E$
 B) $\frac{\pi a^2 E}{\sqrt{2}}$
 C) $\frac{\pi a^2 E}{2\sqrt{2}}$
 D) $\frac{(\pi+2)\pi a^2 E}{(2\sqrt{2})^2}$

Solution : (Correct Answer: B)

We know that.

$$\phi = \oint E dS = E \oint dS \cos 45^\circ$$

In case of hemisphere

$$\phi_{\text{curved}} = \phi_{\text{circular}}$$

$$\text{Therefore, } \phi_{\text{curved}} = E\pi a^2 \cdot \frac{1}{\sqrt{2}} = \frac{E\pi a^2}{\sqrt{2}}$$

40. Two identical charged spheres are suspended by strings of equal lengths. The strings make an angle of 30° with each other. When suspended in a liquid of density 0.8 g cm^{-3} , the angle remains the same. If density of the material of the sphere is 1.6 g cm^{-3} , the dielectric constant of the liquid is

A) 2 B) 1 C) 4 D) 3

Solution : (Correct Answer: A)

From F.B.D of sphere, using Lami's theorem

$$\frac{F}{mg} = \tan \theta$$

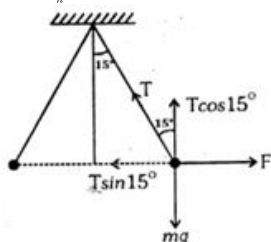
when suspended in liquid, as q remains same,

$$\frac{F'}{mg(1-\frac{\rho}{\sigma})} = \tan \theta$$

$$\frac{F}{mg} = \frac{F'}{mg(1-\frac{\rho}{\sigma})} \text{ where, } F' = \frac{F}{K}$$

$$\frac{F}{mg} = \frac{F'}{mg(1-\frac{\rho}{\sigma})}$$

$$K = \frac{1}{1-\frac{\rho}{\sigma}} = 2$$



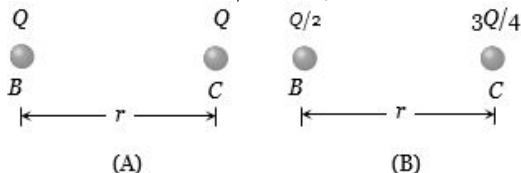
41. Two spherical conductors B and C having equal radii and carrying equal charges in them repel each other with a force F when kept apart at some distance. A third spherical conductor having same radius as that of B but uncharged is brought in contact with B , then brought in contact with C and finally removed away from both. The new force of repulsion between B and C is

A) $F/4$ B) $3F/4$ C) $F/8$ D) $3F/8$

Solution : (Correct Answer: D)

(d) Initially $F = k \cdot \frac{Q^2}{r^2}$ (fig. A). Finally when a third spherical conductor comes in contact alternately with B and C then removed, so charges on B and C are $Q/2$ and $3Q/4$ respectively (fig. B)

$$\text{Now force } F' = k \cdot \frac{(\frac{Q}{2})(\frac{3Q}{4})}{r^2} = \frac{3}{8} F$$



42. Charge on α - particle is

A) $4.8 \times 10^{-19} \text{ C}$ B) $1.6 \times 10^{-19} \text{ C}$
C) $3.2 \times 10^{-19} \text{ C}$ D) $6.4 \times 10^{-19} \text{ C}$

Solution : (Correct Answer: C)

(c) By using $Q = ne \Rightarrow Q = +2e = +3.2 \times 10^{-19} \text{ C}$

43. A body has $-80 \text{ micro coulomb}$ of charge. Number of additional electrons in it will be

A) 8×10^{-5} B) 80×10^{-17}
C) 5×10^{14} D) 1.28×10^{-17}

Solution : (Correct Answer: C)

$$(c) \text{ By } Q = Ne \text{ or } N = \frac{Q}{e} \therefore N = \frac{80 \times 10^{-6}}{1.6 \times 10^{-19}} = 5 \times 10^{14}$$

44. What is the magnitude of a point charge due to which the electric field 30 cm away has the magnitude 2 newton/coulomb [$1/4\pi\epsilon_0 = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$]

A) $2 \times 10^{-11} \text{ coulomb}$ B) $3 \times 10^{-11} \text{ coulomb}$
C) $5 \times 10^{-11} \text{ coulomb}$ D) $9 \times 10^{-11} \text{ coulomb}$

Solution : (Correct Answer: A)

$$(a) \text{ Electric field due to a point charge } E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$q = E \times 4\pi\epsilon_0 r^2 = 2 \times \frac{1}{9 \times 10^9} \times \left(\frac{30}{100}\right)^2 = 2 \times 10^{-11} \text{ coulomb}$$

45. A body can be negatively charged by

A) Giving excess of electrons to it
B) Removing some electrons from it
C) Giving some protons to it
D) चालक पर आवेश एवं उसके क्षेत्रफल दोनों के व्युत्क्रमानुपाती होता है

Solution : (Correct Answer: A)

(a) Excess of electron gives the negative charge on body.

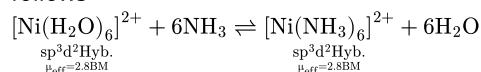
Chemistry

46. Aqueous solution of Ni^{2+} contains $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ and its magnetic moment is 2.83 BM . When ammonia is added in it, comment on the magnetic moment of solution

A) It will remain same
B) It increases from 2.83 BM
C) It decreases from 2.83 BM
D) It cannot be predicted theoretically

Solution : (Correct Answer: A)

On adding ammonia in aqueous solution of Ni^{2+} containing $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ a new complex is formed as follows



As there is no change in hybridisation and number of unpaired electrons so magnetic moment remains same.

47. Which of the following ligand gives chelate complexes?

A) SCN^- B) $\text{C}_2\text{O}_4^{2-}$ C) Pyridine D) $\text{NH}_2 - \text{NH}_3^+$

Solution : (Correct Answer: B)

$\text{C}_2\text{O}_4^{2-}$ is a symmetrical bidentate ligand so it can be used for chelate complex formation.

Pyridine, $\text{NH}_2 - \text{NH}_3^+$ are monodentate ligands so can not form chelate complex.

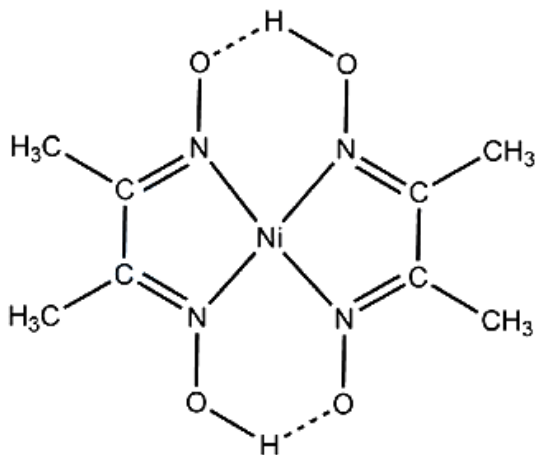
SCN^- is an ambidentate ligand so only one site can be used at a time so no chelate complex is possible.

48. What is the coordination number of Ni in nickel DMG complex?

A) 2 B) 3 C) 6 D) 4

Solution : (Correct Answer: D)

It is evident from the structure of Ni-dmg complex that coordination number of Ni is 4 dmg (dimethyl glyoximate) is a bidentate ligand..



$[\text{Ni}(\text{dmg})_2]$ dmg is bidentate ligand.

49. Which of the following complex compound is "Pentaaquacyanidoiron (III) trichloridotricyanido cobaltate (III)"?

- A) $[\text{Fe}(\text{CN})(\text{H}_2\text{O})_5][\text{CoCl}_3(\text{CN})_3]$
 B) $[\text{Fe}(\text{CN})(\text{H}_2\text{O})_5][\text{CoCl}_3(\text{CN})_3]_3$
 C) $[\text{Fe}(\text{CN})_2(\text{H}_2\text{O})_4]_3[\text{FeCl}_3(\text{CN})_3]_2$
 D) $[\text{Fe}(\text{CN})(\text{H}_2\text{O})_5]_3[\text{CoCl}_3(\text{CN})_3]_2$

Solution : (Correct Answer: D)

In "Pentaaquacyanido Iron (III) trichloridotricyanido cobaltate (III)"

Complex cation is $[\text{Fe}^{\text{III}}(\text{CN})(\text{H}_2\text{O})_5]^{2+}$ and complex anion is $[\text{Co}^{\text{III}}\text{Cl}_3(\text{CN})_3]^{3-}$

Therefore complex is $[\text{Fe}(\text{CN})(\text{H}_2\text{O})_5]_3[\text{CoCl}_3(\text{CN})_3]_2$

50. $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ is a

- A) mixed salt B) double salt
 C) basic salt D) complex salt

Solution : (Correct Answer: B)

It is a double salt as it ionises in aqueous solution as $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O} \rightleftharpoons \text{K}^+ + \text{Mg}^{2+} + 3\text{Cl}^- + 6\text{H}_2\text{O}$

51. Select the compound having maximum conductivity in aqueous medium.

- A) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ B) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
 C) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ D) $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$

Solution : (Correct Answer: A)

As more the number of ion more is conductivity so here $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ shows maximum conductivity as it has four ions.

$[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3 \rightarrow [\text{Cr}(\text{NH}_3)_6]^{3+} + 3\text{Cl}^-$

$[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 \rightarrow [\text{Cr}(\text{NH}_3)_5\text{Cl}]^{2+} + 2\text{Cl}^-$

$[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \rightarrow [\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+ + \text{Cl}^-$

$[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ Does not ionize

52. Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in the alkaline solutions but not in acidic solutions. What is the reason for it?

- A) In acidic solutions, protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available
 B) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
 C) Copper hydroxide is an amphoteric substance
 D) In acidic solutions hydration protects copper ions

Solution : (Correct Answer: A)

Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in the alkaline solutions but not in acidic solutions as in acidic solutions, NH_3 forms a bond with H^+ to give NH_4^+ ion which does not have a lone pair on N to act as a ligand.

53. Which one of the following is an inner orbital complex as well as diamagnetic in behaviour?

- A) $[\text{Zn}(\text{NH}_3)_6]^{2+}$ B) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 C) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ D) $[\text{Co}(\text{NH}_3)_6]^{3+}$

Solution : (Correct Answer: D)

$[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic ($t_{2g}^6 e_g^0$) and is inner orbited complex also as inner 3d electrons get paired here.

$[\text{Zn}(\text{NH}_3)_6]^{2+}$ is outer orbital complex and diamagnetic

$[\text{Cr}(\text{NH}_3)_6]^{3+}$ is inner orbital complex and paramagnetic

54. Which one of the following cyano complexes would exhibit the lowest value of paramagnetic behaviour?

(Atomic no. Cr=24, Mn=25, Fe=26, Co=27)

- A) $[\text{Co}(\text{CN})_6]^{3-}$ B) $[\text{Fe}(\text{CN})_6]^{3-}$ C) $[\text{Mn}(\text{CN})_6]^{3-}$ D) $[\text{Cr}(\text{CN})_6]^{3-}$

Solution : (Correct Answer: A)

	Hybridization	Unpaired electrons	Magnetic moment
$[\text{Co}(\text{CN})_6]^{3-}$	$d^2 sp^3$	0	0
$[\text{Fe}(\text{CN})_6]^{3-}$	$d^2 sp^3$	1	$\sqrt{3}$ BM
$[\text{Mn}(\text{CN})_6]^{3-}$	$d^2 sp^3$	2	$\sqrt{8}$ BM
$[\text{Cr}(\text{CN})_6]^{3-}$	$d^2 sp^3$	3	$\sqrt{15}$ BM

55. Among the following, which are ambidentate ligands?

- (i) SCN^-
 (ii) NO_3^-
 (iii) NO_2^-
 (iv) $\text{C}_2\text{O}_4^{2-}$

- A) (i) and (iii) B) (i) and (iv)
 C) (ii) and (iii) D) (ii) and (iv)

Solution : (Correct Answer: A)

SCN^- and NO_2^- are ambidentate ligands since they have more than one donor atoms to attach to the central metal atom.

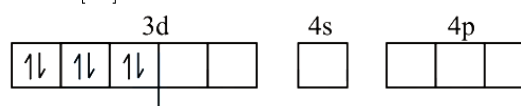
56. Which of the following statement is not correct?

- A) $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$ have the same magnetic moment
 B) $[\text{NiCl}_4]^{2-}$ and $[\text{PtCl}_4]^{2-}$ have different shapes
 C) Hybridisation state of Co in $[\text{Co}(\text{Ox})_3]^{3-}$ is $sp^3 d^2$
 D) In brown-ring complex $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ oxidation state of Fe is +1

Solution : (Correct Answer: C)

$[\text{Co}(\text{Ox})_3]^{3-}$

$\text{Co}^{3+} = [\text{Ar}] 4s^0 3d^6$



$d^2 sp^3$ hybridisation

$[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{Ni}(\text{CO})_4]$ have zero magnetic

moment as have no unpaired electron.
 $[\text{NiCl}_4]^{2-}$ and $[\text{PtCl}_4]^{2-}$ have different shapes that is tetrahedral and square planar respectively.
 In brown-ring complex $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}] \text{SO}_4$ oxidation state of Fe is +1 as NO is taken as +1 here.

57.

Which of the following properties don't help in differentiating, different hydrated isomers of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$?

- A) Conductivity measurements
- B) Precipitation by AgNO_3
- C) Dipole moment
- D) Magnetic moment

Solution : (Correct Answer: D)

Different hydrated

(4) Hydrated isomers of $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ are :

- (i) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
- (ii) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
- (iii) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$

They all have different conductivity, dipole moment and precipitation by AgNO_3 , They all have similar magnetic moment because in all complexes, ligands are weak field. Cr^{+3} has 3 unpaired e^-

58. Which isomerism shown by the compound $[\text{Cr}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$

- A) Co-ordination
- B) Geometrical
- C) Optical
- D) Linkage

Solution : (Correct Answer: D)

NO_2 is ambident and can be linked either to N— side as $(-\text{NO}_2)$ or to O— side as $(-\text{ONO})$.

59. Which of the following species must have maximum number of electrons in ' d_{xy} ' orbital?

- A) Cr
- B) Fe^{3+}
- C) Cu^+
- D) Both A and B

Solution : (Correct Answer: C)

A. $\text{Cr} \rightarrow 4s^1 3d^5$ (each d-orbital has $1e^-$)

B. $\text{Fe}^{3+} \rightarrow 4s^0 3d^5$ (each d-orbital has $1e^-$)

C. $\text{Cu}^+ \rightarrow 4s^0 3d^{10}$ (each d-orbital has $2e^-$)

60. In an octahedral structure, the pair of d-orbitals involved in $d^2 sp^3$ hybridisation is

- A) $d_{x^2-y^2}, d_{z^2}$
- B) $d_{xz}, d_{x^2-y^2}$
- C) d_{z^2}, d_{xz}
- D) d_{xy}, d_{yz}

Solution : (Correct Answer: A)

In the formation of $d^2 sp^3$ hybrid orbitals, two $(n-1)d$ orbitals of e_g set i.e., $(n-1) d_{z^2}$ and $(n-1) d_{x^2-y^2}$ orbitals, one ns and three np (np_x , np_y and np_z) orbitals combine together and form six $d^2 sp^3$ hybrid orbitals.

61. The crystals of ferrous sulphate on heating give

- A) $\text{FeO} + \text{SO}_2 + \text{H}_2\text{O}$
- B) $\text{Fe}_2\text{O}_3 + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
- C) $\text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$
- D) $\text{FeO} + \text{SO}_3 + \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$

Solution : (Correct Answer: C)

$\text{FeSO}_4 \cdot 7\text{H}_2\text{O} \xrightarrow{\Delta} \text{FeSO}_4 + 7\text{H}_2\text{O}$

$2\text{FeSO}_4 \xrightarrow{\Delta} \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$

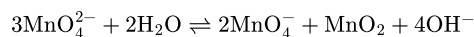
62. Which statements are incorrect about transition elements?

- A) They show variable valency
- B) They readily form complex compounds
- C) All their ions are colourless
- D) Their ions contain partially filled d-orbital

Solution : (Correct Answer: C)

"All their ions are colourless" this is incorrect because about 90% of ions are coloured and only few are colourless.

63. KMnO_4 can be prepared from K_2MnO_4 as per the reaction



The reaction can go to completion by removing OH^- ions by adding

- A) KOH
- B) CO_2
- C) SO_2
- D) HCl

Solution : (Correct Answer: B)

As MnO_4^- can oxidise SO_2 , HCl but not CO_2 as in CO_2 C-atom has its highest oxidation state (+4) it means CO_2 can be used here.

64. The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence:

- A) $\text{Ga} < \text{In} < \text{Al} < \text{Tl}$
- B) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
- C) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$
- D) $\text{In} < \text{Tl} < \text{Ga} < \text{Al}$

Solution : (Correct Answer: B)

Stability of +1 oxidation state due to inert pair effect is in the order $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$

65. The most basic hydroxide from the following is

- A) $\text{Pr}(\text{OH})_3$ (Z = 59)
- B) $\text{Sm}(\text{OH})_3$ (Z = 62)
- C) $\text{Ho}(\text{OH})_3$ (Z = 67)
- D) $\text{La}(\text{OH})_3$ (Z = 57)

Solution : (Correct Answer: D)

The basic nature of hydroxides decreases from La to Lu along the lanthanide series due to lanthanoid contraction. Thus, $\text{La}(\text{OH})_3$ is the most basic hydroxide.

66. The most convenient method to protect the bottom of ship made of iron is

- A) coating it with red lead oxide
- B) white tin plating
- C) connecting it with Mg block
- D) connecting it with Pb block

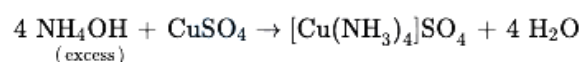
Solution : (Correct Answer: B)

The most convenient method to protect the bottom of the ship made of iron is white tin plating preventing the build up of barnacles.

67. When to a copper sulphate solution, excess of ammonium hydroxide added then

- A) No change is observed
- B) Blue precipitate of copper hydroxide is obtained
- C) A deep blue solution is obtained
- D) Black precipitate of copper oxide is obtained

Solution : (Correct Answer: C)



68. Which of the following statements is not correct about the electronic configuration of gaseous chromium atom
- A) It has 5 electrons in 3d and one electron in 4s orbitals
 B) The principal quantum numbers of its valence electrons are 3 and 4
 C) It has 6 electrons in 3d orbital
 D) Its valence electrons have quantum number $l = 0$

Solution : (Correct Answer: C)

Chromium has atomic number of 24, its electronic configuration is $1s^2 2s^2 2p^6 3s^2 3d^5 4s^1$

69. Which of the following ions will liberate iodine when treated with KI?

A) Cu^{2+} B) Fe^{2+} C) Pb^{2+} D) Sn^{2+}

Solution : (Correct Answer: A)

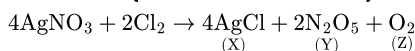
Cu^{2+} ions can oxidize iodide to iodine.

70. $4\text{AgNO}_3 + 2\text{Cl}_2 \rightarrow 4\text{X} + 2\text{Y} + \text{Z}$
 (dry)

In the above sequence of reaction y and z are respectively

A) NO_2 and N_2O_4 B) N_2O_5 , O_2
 C) AgCl and O_2 D) AgCl , N_2

Solution : (Correct Answer: B)



Hence Y and Z are N_2O_5 and O_2 respectively.

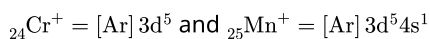
71. Among the following transition elements, which one has the highest second ionisation energy?

(i) V ($Z = 23$) (ii) Cr ($Z = 24$) (iii) Mn ($Z = 25$)

A) (i) B) (ii) C) (iii) D) (iv)

Solution : (Correct Answer: B)

The general trend observed in the second ionisation enthalpy of 3d-series elements is that as we move from left to right in the period, the ionisation energy increases. But, Cr and Zn are exceptions to this trend. The value of second ionisation energy for Cr is more than Mn. This can be explained based on the electronic configuration of their M^+ ions:



Cr^+ has a stable electronic configuration because of half-filled d-subshell. Thus removing an electron from it is difficult. Therefore, Cr has the highest second ionisation energy among the given elements.

The correct order for second ionisation energy for the given elements is $\text{Ti} < \text{V} < \text{Mn} < \text{Cr}$.

72. Which one of the following has strongest metallic bonding?

A) Fe B) Sc C) V D) Cr

Solution : (Correct Answer: D)

Strength of metallic bond depends upon number of unpaired electrons. As number of unpaired electrons increase, the bond strength also increases. So, Cr, Mo, show stronger bonding due to maximum number of unpaired electrons.

73. Increasing basic properties of TiO_2 , ZrO_2 and HfO_2 are in order:

A) $\text{TiO}_2 < \text{ZrO}_2 < \text{HfO}_2$
 B) $\text{HfO}_2 < \text{ZrO}_2 < \text{TiO}_2$
 C) $\text{HfO}_2 < \text{TiO}_2 < \text{ZrO}_2$
 D) $\text{ZrO}_2 < \text{TiO}_2 < \text{HfO}_2$

Solution : (Correct Answer: A)

Basicity of metal oxides in d-block element is increase down the group.

74. Lanthanide contraction is caused due to

A) The imperfect shielding on outer electrons by 4 f-electrons from the nuclear charge
 B) The appreciable shielding on outer electrons by 4 f-electrons from the nuclear charge
 C) The appreciable shielding on outer electrons by 5d-electrons from nuclear charge
 D) The same effective nuclear charge from Ce to Lu

Solution : (Correct Answer: A)

Lanthanide contraction is caused due to poor screening effect of 4f orbitals

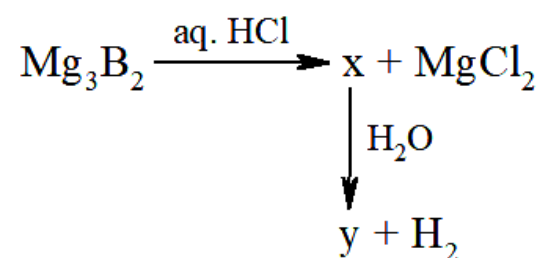
75. Aqua-regia reacts with Pt to yield –

A) $\text{Pt}(\text{NO}_3)_4$ B) H_2PtCl_6 C) PtCl_4 D) PtCl_2

Solution : (Correct Answer: B)

Aqua-regia is a mixture of nitric acid and hydrochloric acid in molar ratio 1 : 3. It is yellow orange fuming liquid. In Aqua-regia Pt form chloroplatinic acid, H_2PtCl_6

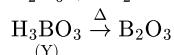
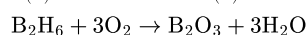
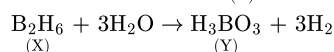
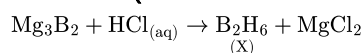
- 76.



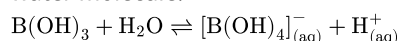
for (X) and (Y) the incorrect choice is?

A) (X) is BCl_3
 B) (Y) is H_3BO_3
 C) (X) with air and (Y) on strong heating (red heat) give same compound
 D) In (Y) boron completes its octet by moving H^+ from water molecule

Solution : (Correct Answer: A)



In Y, B completes its octet by removing H^+ from water molecule.



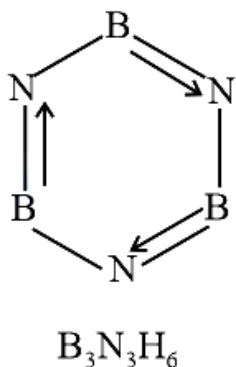
- 77.

Identify the correct statements about borazene, $\text{B}_3\text{N}_3\text{H}_6$:

- (i) Borazene is aromatic.
 (ii) There are four isomers of bi substituted molecule of borazene molecules, $(B_3N_3H_4X_2)$.
 (iii) Borazene is more reactive towards addition reactions than benzene.

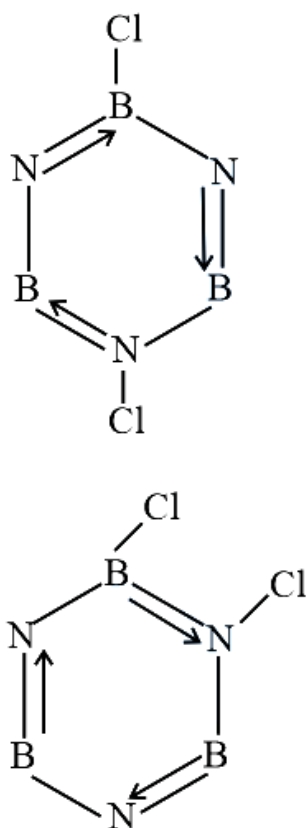
- A) Only (i)
 B) (i) and (ii)
 C) (i) and (iii)
 D) (i), (ii) and (iii)

Solution : (Correct Answer: D)



it is a aromatic compound like benzene and due to electro negativeity difference it is polar and more reactive than benzene

Possible of isomers $B_3N_3H_6$:-



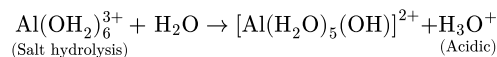
- (i), (ii) and (iii)

78. The aqueous solution of $AlCl_3$ is acidic due to the hydrolysis of

- A) aluminium ion
 B) chloride ion
 C) both aluminium and chloride ions
 D) none of the above

Solution : (Correct Answer: A)

The aqueous solution of $AlCl_3$ is acidic due to the hydrolysis of aluminium ion.



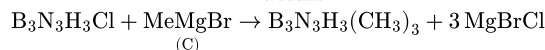
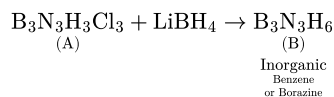
Cl^- does not hydrolyse.

79. The reaction of $H_3N_3B_3Cl_3$ (A) with $LiBH_4$ in tetrahydrofuran gives inorganic benzene (B). Further, the reaction of (A) with (C) leads to $H_3N_3B_3(Me)_3$. Compounds (B) and (C) respectively, are

- A) Borazine and MeBr
 B) Boron nitride and MeBr
 C) Diborane and MeMgBr
 D) Borazine and MeMgBr

Solution : (Correct Answer: D)

Here B is Borazine and C is MeMgBr. The reactions are as follows



80. B has a smaller first ionization enthalpy than Be. Consider the following statements

- (i) it is easier to remove 2p electron than 2s electron
 (ii) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be
 (iii) 2s electron has more penetration power than 2p electron
 (iv) atomic radius of B is more than Be (atomic number B = 5, Be = 4)

The correct statements are

- A) (i), (ii) and (iv)
 B) (i), (iii) and (iv)
 C) (i), (ii) and (iii)
 D) (ii), (iii) and (iv)

Solution : (Correct Answer: C)

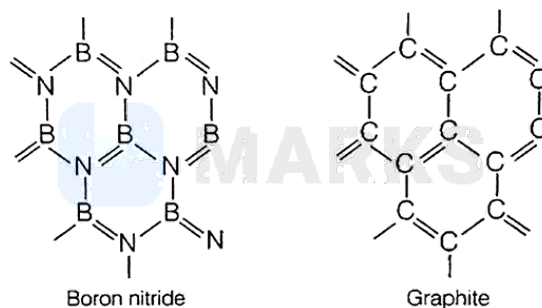
As radius of Boron is less than radius of Be hence IV statement is false while rest other are true statements.

81. Which of the following structure is similar to graphite?

- A) B $\quad$ B) B_4C $\quad$ C) B_2H_6 $\quad$ D) BN

Solution : (Correct Answer: D)

$(BN)_x$ is inorganic graphite with graphite like structure.



Boron, B₂H₆ (Diborane) and B₄C (Norbide) has different structures than graphite.

82. Which statements is not correct about H₃BO₃?

- A) It is a strong tribasic acid
- B) It has a layer structure in which planar BO₃ units are joined by hydrogen bonds
- C) It is prepared by acidifying an aqueous solution of borax
- D) It does not act as proton donor but acts as a Lewis acid by accepting hydroxyl ion

Solution : (Correct Answer: A)

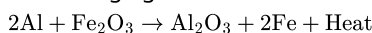
H₃BO₃ is orthoboric acid, it is weak monobasic Lewis acid of boron. It has a layer structure in which planar BO₃ units are joined by hydrogen bonds.

83. In the aluminothermite process, aluminium acts as

- A) an oxidizing agent B) a flux
- C) a reducing agent D) a solder

Solution : (Correct Answer: C)

In the aluminothermite process, aluminium acts as a reducing agent.



Al reduces Fe₂O₃ to Fe.

84. When orthoboric acid is heated strongly, it gives

- A) B₂O₃ B) H₂B₄O₇ C) HBO₂ D) B

Solution : (Correct Answer: A)

When orthoboric acid is heated strongly, it gives



85. During Hoopes process of electrolytic refining of Al, the middle layer is of

- A) Pure Al B) Impure Al
- C) Cryolite + BaF₂ D) Alloys of Al, Ca, Si

Solution : (Correct Answer: C)

In the Hoopes process for the purification of aluminium, the middle layer consists of cryolite and BaF₂.

86. B₂H₆ + NH₃ → Addition compound (X) $\xrightarrow{450\text{ K}}$ Y + Z (g)
In the above sequence Y and Z are respectively

- A) Borazine, H₂ B) Boron, H₂
- C) Boron nitride, H₂ D) Boron, Hydrazine

Solution : (Correct Answer: A)

Here the reaction sequence is as follows



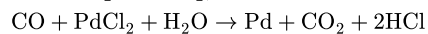
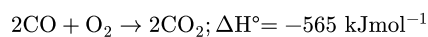
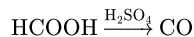
Hence, Y is borazine and Z is hydrogen.

87. Which of the following statement is correct about CO?

- A) It reduces aqueous solution of PdCl₂ to metallic Pd
- B) CO is neutral oxide and acts as a fuel
- C) In laboratory it is prepared by dehydrating HCOOH with conc. H₂SO₄
- D) All are correct

Solution : (Correct Answer: D)

Here all statements regarding CO are correct as follows

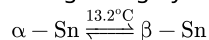


88. The soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White metallic tin buttons got converted to grey powder. This transformation is related to

- A) an interaction with nitrogen of the air at very low temperatures
- B) a change in the crystalline structure of tin
- C) an interaction with water vapour contained in the humid air
- D) a change in the partial pressure of oxygen in the air.

Solution : (Correct Answer: B)

As temperature decreases, white tin (β – form) changes to grey tin (α – form)



α – Sn has a much lower density than β – Sn.

89. Aluminium chloride exists as dimer, Al₂Cl₆ in solid state as well as in solution: of non-polar solvents such as benzene. when dissolved in water, it gives

- A) Al₂O₃ + 6HCl B) Al³⁺ + 3Cl[–]
- C) [Al(OH)₆]^{3–} + 3HCl D) [Al(H₂O)₆]³⁺ + 3Cl[–]

Solution : (Correct Answer: D)

AlCl₃ is covalent but in water, it become ionic due to large hydration energy it dissociated as follows

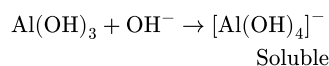
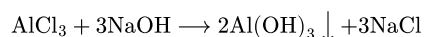


90. Aqueous ammonia is used as a precipitating reagent for Al³⁺ ions as Al(OH)₃ rather than aqueous NaOH, because

- A) NH₄⁺ is a weak base
- B) NaOH is a very strong base
- C) NaOH forms [Al(OH)₄][–] ions
- D) NaOH forms [Al(OH)₂]⁺ ions.

Solution : (Correct Answer: C)

Al(OH)₃ formed with NaOH dissolves in excess of NaOH to form aluminate ion, which is stable.



Biology - (Zoology)

91. Who proposed that the genetic code for amino acids should be made up of three nucleotides?

- A) George Gamow B) Francis Crick
- C) Jacques Monod D) Franklin Stahl

Solution : (Correct Answer: A)

George Gamow, a physicist proposed that genetic code for amino acids should be made up of three nucleotides.

92. Given below are two statements :

Statement I : Transfer RNAs and ribosomal RNA do not interact with mRNA.

Statement II : RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

In the light of the above statements, choose the most appropriate answer from the options given below :

- A) Both statement I and statement II are correct
- B) Both statement I and statement II are incorrect
- C) Statement I is correct but statement II is incorrect
- D) Statement I is incorrect but statement II is correct

Solution : (Correct Answer: D)

Both transfer RNAs and ribosomal RNA interact with mRNA.

RNA interference (RNAi) takes place in all eukaryotic organisms as a method of cellular defence.

93. Match List I with List II:

	List - I		List - II
A.	Alfred Hershey and Martha Chase	(i)	Streptococcus pneumoniae
B.	Euchromatin	(ii)	Densely packed and dark-stained
C.	Frederick Griffith	(iii)	Loosely packed and light-stained
D.	Heterochromatin	(iv)	DNA as genetic material as a genetic

Choose the correct answer from the options given below:

- A) A-II, B-IV, C-I, D-III
- B) A-IV, B-II, C-I, D-III
- C) A-IV, B-III, C-I, D-II
- D) A-III, B-II, C-IV, D-I

Solution : (Correct Answer: C)

The unequivocal proof that DNA is the genetic material came from the experiment of Alfred Hershey and Martha Chase.

Euchromatin are lightly stained region with loosely packed chromatin fibre.

Frederick Griffith performed series of experiments by selecting the different strains of Streptococcus pneumoniae.

Heterochromatin are darkly stained region with tightly packed chromatin fibre.

94. Given below are two statements :

Statement I : In the RNA world, RNA is considered the first genetic material evolved to carry out essential life processes. RNA acts as a genetic material and also as a catalyst for some important biochemical reactions in living systems. Being reactive, RNA is unstable.

Statement II: DNA evolved from RNA and is a more stable genetic material. Its double helical strands being complementary, resist changes by evolving repairing mechanism.

In the light of the above statements, choose the most appropriate answer from the options given below :

- A) Both statement I and statement II are correct
- B) Both statement I and statement II are incorrect
- C) Statement I is correct but statement II is incorrect
- D) Statement I is incorrect but statement II is correct

Solution : (Correct Answer: A)

In RNA world, RNA was the first genetic material as there are enough evidences to suggest that essential life processes (such as metabolism, translation, splicing, etc) evolved around RNA. RNA used to act as a genetic material as well as catalyst (there are some important biochemical reaction in living systems that are catalysed by RNA catalysts not by protein enzymes) so, statement I is correct statement II is also correct as DNA being double stranded and having complementary strands further resists changes by evolving a process of repair.

95. Which of the following are the post-transcriptional events in an eukaryotic cell?

- A. Transport of pre-mRNA to cytoplasm prior to splicing.
- B. Removal of introns and joining of exons.
- C. Addition of methyl group at 5' end of hnRNA.
- D. Addition of adenine residues at 3' end of hnRNA.
- E. Base pairing of two complementary RNAs.

Choose the correct answer from the options given below :

- A) A, B, C only
- B) B, C, D only
- C) B, C, E only
- D) C, D, E only

Solution : (Correct Answer: B)

The process of copying genetic information from one strand of the DNA into RNA is known as transcription. It occurs in the cytoplasm with the help of transcribing enzyme.

Transport of pre-mRNA to cytoplasm prior to splicing is a part of transcription.

The primary transcript is converted into functional mRNA after post transcriptional processing involves 3 steps as follows-

- Modification of 5' end by capping,
- Tailing,

- Splicing.

Base pairing of two complementary RNA is not an event of post-transcription. Hence, statements B, C, D are post-transcriptional modification events in eukaryotic cell.

96. Which factor is important for termination of transcription?

A) α (alpha) B) σ (sigma)
C) ρ (rho) D) γ (gamma)

Solution : (Correct Answer: C)

In prokaryotes the RNA polymerase is only capable of catalysing the process of elongation. It associates transiently with initiation factor (σ) and termination factor (ρ) to initiate and terminate the transcription respectively.

97. Twins are born to a family that lives next door to you. The twins are a boy and a girl. Which of the following must be true?

A) They are monozygotic twins.
B) They are fraternal twins.
C) They were conceived through in vitro fertilization.
D) They have 75% identical genetic content.

Solution : (Correct Answer: B)

Fraternal twins or dizygotic twins are 2 separate fertilized eggs, they usually develop 2 separate amniotic sacs, placentas and supporting structures.

If twins are a boy and a girl, this indicates they are fraternal twins.

98. Which chromosome in the human genome has the highest number of genes?

A) Chromosome X B) Chromosome Y
C) Chromosome 1 D) Chromosome 10

Solution : (Correct Answer: C)

In human genome, Chromosome 1 has the highest number of genes, i.e., 2968.

99. Histones are enriched with -

A) Lysine & Arginine
B) Leucine & Lysine
C) Phenylalanine & Leucine
D) Phenylalanine & Arginine

Solution : (Correct Answer: A)

In eukaryotes, packaging of DNA is much more complex. There is a set of positively charged, basic proteins called histones.

Histones are organised to form a unit of light molecules called histone octamer.

They are rich in the basic amino acid residues lysine and arginine.

100. Which of the following statements are correct about Klinefelter's Syndrome?

A. This disorder was first described by Langdon Down (1866).
B. Such an individual has overall masculine development. However, the feminine development is also expressed.
C. The affected individual is short statured.

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile.

Choose the correct answer from the options given below:

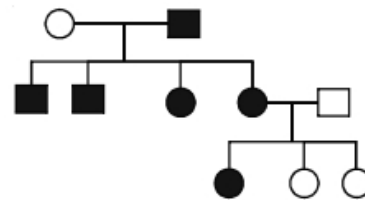
A) A and E only
B) A and B only
C) C and D only
D) B and E only

Solution : (Correct Answer: D)

Klinefelter's syndrome is caused due to the presence of an additional copy of X-chromosome resulting into a karyotype of 47, XXY. Such an individual has overall masculine development, however, the feminine development is also expressed. Such individuals are sterile. Thus, statement B and E are correct regarding Klinefelter's syndrome.

Statement A, C and D are incorrect w.r.t. Klinefelter's syndrome as they are associated with Down's syndrome.

101. In the given pedigree, indicate whether the shaded symbols indicate dominant or recessive allele.



A) Recessive
B) Codominant
C) Dominant
D) It can be recessive or dominant both

Solution : (Correct Answer: D)

If propositus $\phi \rightarrow$ is Aa

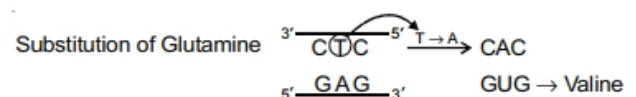
Then it will be dominant pedigree

If $\rightarrow o'$ is aa $\rightarrow$ then it will be recessive pedigree.

102. Sickle cell-anaemia disorder arises due to

A) Duplication of a segment of DNA
B) Substitution in a single base of DNA
C) Deletion of a segment of DNA
D) Duplication in a base pair of RNA

Solution : (Correct Answer: B)



103. Down's syndrome is an example of

A) Aneuploidy B) Polyteny
C) Polyploidy D) Monoploidy

Solution : (Correct Answer: A)

Aneuploidy is an abnormal number of chromosomes and is a type of chromosome abnormality. The presence of an extra chromosome 21 is found in Down's syndrome.

104. Which of these is not a Mendelian disorder?

- A) Cystic fibrosis B) Sickle-cell anaemia
C) Colourblindness D) Turner's syndrome

Solution : (Correct Answer: D)

The disorder cystic fibrosis, sickle cell anaemia, colour blindness and haemophilia are caused due to the abnormality in gene (Mendelian factor).

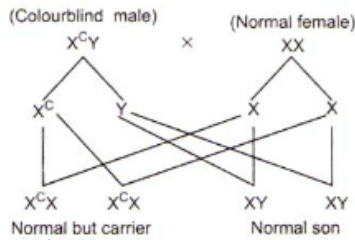
Turner's syndrome is due to chromosomal abnormality. It is characterized by 44 autosomes and only one X-chromosome.

105. If a man who is colour blind marries a woman, who is pure normal for colour vision, the chances of their sons have colour blindness is

- A) 100% B) 50 : 50
C) 0% D) 75 : 25

Solution : (Correct Answer: C)

So, all sons in the progeny will be normal.



106. Dihybrid ratio of test cross 1 : 1 : 1 : 1 proves that

- A) F_1 hybrid produces four different progenies
B) F_1 hybrid produces two different progenies
C) Parents produce two different progenies
D) None of the above

Solution : (Correct Answer: A)

Dihybrid cross is a cross involving two pairs of contrasting characters. A dihybrid test cross gives 1 : 1 : 1 : 1 ratio indicating that when F_1 hybrid is crossed with recessive parent the two pairs of factors segregate and assorting independently and produce four type of progenies.

107. Mendel obtained recessive character in F_2 by ...
A... the ...B... plants. Here A and B refers to

- A) A—self-pollinating; B — F_1
B) A—self-pollinating; B — F_2
C) A—cross-pollinating ; B — F_1
D) A—cross-pollinating; B — F_2

Solution : (Correct Answer: A)

Mendel obtained the recessive character in F_2 by self pollinating the F_1 -plants.

Mendel cross-pollinated a pure tall pea plant (100 – 120 cm high) and a pure dwarf pea plant. (only 22 to 44 cm high). He called them parental generation, expressed now-a-days by symbol P.

This hybridization popularly called as monohybrid cross

[Image]

This three generations of pea plants after crossing a pure tall plant with a dwarf one. The plants of F_1 -generation are all tall, of F_2 generation three tall and one dwarf. One third of the tall plants are pure, while the remaining behave as hybrids

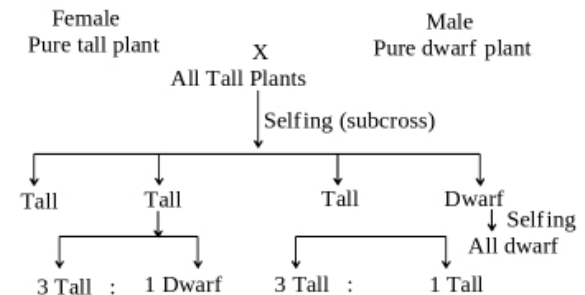
F_1 -generation Seeds collected from the parental generation called first filial generation or F_1 -

generation

F_2 -generation F_1 -plants pollinated among them self (self breeding or inbreeding) and seed produced by F_1 -plants called F_2 -generation. They were in ratio 3 : 1 (three tall and one dwarf).

F_3 -generation Mendel allowed F_2 -plant to form seed by self-pollination called F_3 -generation. Mendel observed that tall and dwarf plant behave differently

- (i) Dwarf plant produced dwarf plant on self-pollinated
(ii) In tall plants one third plants breed true so they were pure
(iii) Other two third plant behave like parents and give tall to dwarf plants 3 : 1 indicate that their parents have dwarf genes also



108. The number of contrasting characters studied by Mendel for his experiments was

- A) 7 B) 14 C) 4 D) 2

Solution : (Correct Answer: A)

Mendel took 7 contrasting characters of pea plant for his experiment.

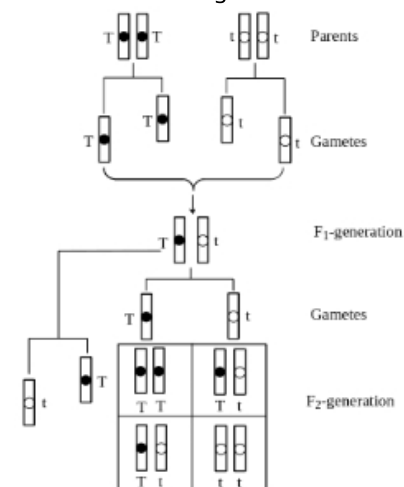
109. Mendel conducted experiments for

- A) 7 years B) 6 years
C) 5 years D) 4 years

Solution : (Correct Answer: A)

Mendel carried out hybridization experiments on garden pea for 7 years from 1856 – 1863

110. What does this diagram indicate?



- A) Law of dominance interpreted on basis of genes
B) Law of segregation interpreted on basis of genes
C) Law of independent assortment interpreted on basis of genes

D) Simply gamete genes

Solution : (Correct Answer: B)

As we can see in the diagram given in question there are only one trait considered. So, this diagram depict law of segregation explained on the behalf of gene in which genes or factors are indicated as solid and yellow situated on two different chromosome

111. A woman has a haemophilic son and three normal children. Her genotype and that of her husband with respect to this gene would be

- A) $XX \wedge X^hY$
 B) $X^hX^h \wedge X^hY$
 C) X^hX^h and XY
 D) $X^hX \wedge XY$

Solution : (Correct Answer: D)

112. Which of the following law was discovered first by Mendel?

- A) Law of dominance
 B) Law of segregation
 C) Law of independent assortment
 D) Law of sex determination

Solution : (Correct Answer: B)

Originally, Mendel proposed two laws, firstly law of segregation and then law of independent assortment. Mendel coined the term dominant for any trait that express itself when present with the factors for the contrasting trait, and used the word recessive for any trait that is not expressed when present alongwith contrasting dominant trait in the hybrid.

113. A : Blood group phenotype is controlled by presence or absence of antigens present on surface coating of RBC.

R : These antigens are of three types and found in the oligosaccharides rich head regions of a glycoprotein

- A) Assertion and Reason both are correct and also correct explanation.
 B) Assertion and Reason both are correct but not explanation of assertion.
 C) Assertion is correct, but Reason is incorrect.
 D) Both Assertion and Reason are incorrect.

Solution : (Correct Answer: B)

114. A : Dominance is not an autonomous feature of a gene.

R : It depends as much on the gene product and the production of a particular phenotype from this product

- A) Assertion and Reason both are correct and also correct explanation.
 B) Assertion and Reason both are correct but not explanation of assertion.
 C) Assertion is correct, but Reason is incorrect.
 D) Both Assertion and Reason are incorrect.

Solution : (Correct Answer: A)

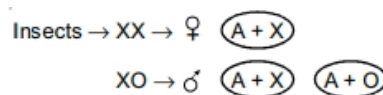
115. While solving the problem of sex determination in large number of insects, it was observed that

- A) All eggs lack sex chromosome
 B) Some of the sperms bear the X-chromosome

C) All eggs as well as sperms bear the X-chromosome

D) Some of the eggs bear the X-chromosome

Solution : (Correct Answer: B)

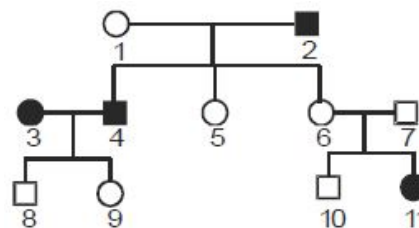


116. Which of the following gametes are produced by individual with genotype $RrYy$?

- A) Ry, RY B) ry, ry
 C) Ry, ry, rY D) Ry, RY, rY, ry

Solution : (Correct Answer: D)

117. In Huntington's disease, the unaffected persons are homozygous for normal allele h . The following is erroneous because



- A) it shows both male and female affected by Huntingtons disease
 B) either person 6 or 7 should have the disease, if individual 11 shows the disease.
 C) at least one of the 2 children (8, 9) should have the disease
 D) all of these

Solution : (Correct Answer: B)

Since the offspring number 11 is affected (diseased) either of the two parents 6 or 7 have to be affected.

118. Loss of a single chromosome produces a condition called.

- A) Haploidy B) Nullisomy C) Trisomy D) Monosomy

Solution : (Correct Answer: D)

119. Which is incorrect in Mendelian characters ?

Character	dominant	recessive
A) Pod colour	green	yellow
B) Seed shape	round	wrinkled
C) Flower position	terminal	Axillary
D) Flower colour	violet	white

Solution : (Correct Answer: C)

120. In a dihybrid cross between $RRYY$ and $rryy$, the number of $RrYy$ F_2 genotypes will be

- A) 4 B) 3 C) 2 D) 9

Solution : (Correct Answer: A)

121. In his classic experiments on pea plants, Mendel did not use

- A) seed shape B) flower position
 C) seed colour D) pod length.

Solution : (Correct Answer: D)

(d) : Mendel considered the following characters of pea in his experiments :

Character	Dominant	Recessive

1	Seed shape	Round (<i>R</i>)	Wrinkled (<i>r</i>)	None of the above
2	Seed cotyledon colour	Yellow (<i>Y</i>)	Green (<i>y</i>)	Solution : (Correct Answer: B)
3	Flower colour	Violet (<i>V</i>)	White (<i>v</i>)	It's Obvious
4	Pod shape	Inflated (<i>I</i>)	Constricted (<i>i</i>)	127. Which one is the sex linked trait
5	Pod colour	Green (<i>G</i>)	Yellow (<i>g</i>)	A) Myxoedema
6	Flower position	Axial (<i>A</i>)	Terminal (<i>a</i>)	C) Blindness
7	Stem height	Tall (<i>T</i>)	Dwarf (<i>t</i>)	Solution : (Correct Answer: D)

122. Between persons of which two blood groups is the blood transfusion not possible

- A) *O* and *AB* (*AB* recipient)
- B) *O* and *A* (*O* donor)
- C) *O* and *B* (*O* donor)
- D) *O* and *AB* (*AB* donor)

Solution : (Correct Answer: D)

(d) The people of blood group *AB* can receive transfusions of blood from people of any of the four blood groups, are called universal recipients.

123. Gene can be defined as

- A) Unit of segregation
- B) Unit of physiological activity
- C) Unit of recombination
- D) Unit of function

Solution : (Correct Answer: C)

It's Obvious

124. *RR* (Red) is crossed with *ww* (white), all the *Rw* offsprings are pink. This is an indication that *R* gene is

- A) Hybrid
- B) Recessive
- C) Incompletely dominant
- D) Mutant

Solution : (Correct Answer: C)

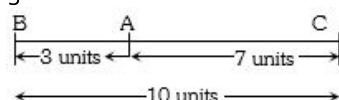
(c) This is an example of incomplete dominance in which the genes from both the parents are accumulated in F_1 which show pink phenotype (This is an exception of Mendel's law).

125. The map distance between genes *A* and *B* is 3 units, between *B* and *C* 10 units and between *C* and *A* 7 units. The order of the genes in a linkage map constructed on the above data would perhaps be

- A) *A, B, C*
- B) *A, C, B*
- C) *B, C, A*
- D) *B, A, C*

Solution : (Correct Answer: D)

(d) From the recombination frequencies between different genes *A, B* and *C*, it is clear that distance between *A* and *B* genes is 3 units, between *B* and *C* 10 units and *C* and *A* 7 units. Hence linear order of genes on chromosomes is :



126. In case of disputed parentage, the blood group analysis of the mother, child and alleged father can

- A) Definitely prove a man to be the father
- B) Only prove that he cannot be the father
- C) Not be of any use

128. Mutation caused by a mutagen is

- A) Natural
- B) Chemical
- C) Spontaneous
- D) Induced

Solution : (Correct Answer: D)

(d) 1) Induced mutations are alterations in the gene after it has come in contact with mutagens and environmental causes. Mutagens can be chemical (nitrous acid), radiations (x-ray) etc. 2) A mutation which occurs by itself without first being affected by a mutagen, for example during the process of DNA replication. 3) Mutation-induced by chemicals such as Nitrous acid is called chemical mutation. 4) Spontaneous mutations are the mutations that are a result of natural changes in DNA structure.

129. Pod character i.e. green colour in pea is

- A) Dominant
- B) Incompletely dominant
- C) Recessive character
- D) Abnormal character

Solution : (Correct Answer: A)

(a) Yellow colour in pea is recessive character of pod colour.

130. Which of the following has been used for genetic researches

- A) *Pisum*
- B) *Neurospora*
- C) *E. coli*
- D) All the above

Solution : (Correct Answer: D)

It's Obvious

131. If in a garden pea plant, a cross is made between red flowered and white flowered plants. What will be the phenotypic ratio in F_2 generation

- A) 1 : 2 : 1
- B) 9 : 3 : 3
- C) 3 : 1
- D) 1 : 3

Solution : (Correct Answer: C)

(c) $\frac{RR}{\text{(Red flower)}} \times \frac{rr}{\text{(White flower)}} \rightarrow F_1 = Rr$

<i>R</i>	<i>r</i>
<i>R</i> <i>RR</i>	<i>Rr</i>
<i>r</i> <i>rR</i>	<i>rr</i>

After selfing

RR = Homozygous for red flower

Rr and *rR* = Heterozygous for red flower

rr = homozygous for white flower

3 Red and 1 White = 3 : 1

132. When a heterozygous dominant is crossed with homozygous recessive than ratio of progeny will be

- A) 1 : 2 B) 2 : 1 C) 3 : 1 D) 1 : 1

Solution : (Correct Answer: D)

(d) It is an example of test cross.

133. Mendel chose pea plants because

- A) They were cheap
 B) They were having seven pairs of contrasting characters
 C) They were easily available
 D) Of great economic importance

Solution : (Correct Answer: B)

(b) Mendel selected these 7 characters -

- (1) Stem length (2) Flower position
 (3) Pod shape (4) Pod colour
 (5) Seed shape (6) Seed colour
 (7) Seed coat colour

134. If a cross is made between AA and aa , the nature of F_1 progeny will be

- A) Genotypically AA , phenotypically a
 B) Genotypically Aa , phenotypically a
 C) Genotypically Aa , phenotypically A
 D) Genotypically aa , phenotypically A

Solution : (Correct Answer: C)

(c) $AA \times aa - P$
 $Aa - F_1$

135. The branch of botany dealing with heredity and variation is called

- A) Geobotany B) Sericulture
 C) Genetics D) Evolution

Solution : (Correct Answer: C)

(c) Genetics is the study of principles and mechanism of heredity and variations.

Biology - (Botany)

136. Read the following statements and choose the set of correct statements:

- (a) Euchromatin is loosely packed chromatin
 (b) Heterochromatin is transcriptionally active
 (c) Histone octamer is wrapped by negatively charged DNA in nucleosome
 (d) Histones are rich in lysine and arginine
 (e) A typical nucleosome contains 400 bp of DNA helix

Choose the correct answer from the options given below:

- A) (a), (c), (d) Only
 B) (b), (e) Only
 C) (a), (c), (e) Only
 D) (b), (d), (e) Only

Solution : (Correct Answer: A)

137. Which is the "Only enzyme" that has "Capability" to catalyse Initiation, Elongation and Termination in the process of transcription in prokaryotes?

- A) DNA dependent DNA polymerase

- B) DNA dependent RNA polymerase
 C) DNA Ligase
 D) DNase

Solution : (Correct Answer: B)

138. Matching sequence of DNA between two evidences, one of the criminal with the suspect is known as:

- A) DNA fingerprinting
 B) DNA amplification
 C) Gene mapping
 D) DNA resolution

Solution : (Correct Answer: A)

139. Given below are two statements:

Statement I: In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II: In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the correct answer from the options given below:

- A) Statement I is incorrect but Statement II is true.
 B) Both Statement I and Statement II are true.
 C) Both Statement I and Statement II are false.
 D) Statement I is correct but Statement II is false.

Solution : (Correct Answer: A)

In prokaryotes, the negatively charged DNA is held with some positively charged proteins in a region termed as nucleoid.

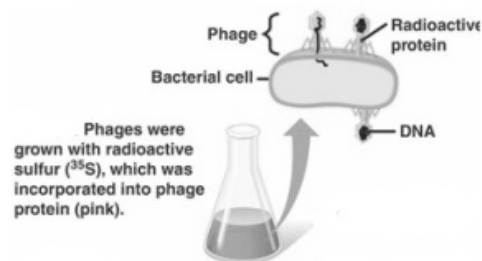
In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome.

140. A bacterium grown over medium having radioactive Sulphur incorporates radioactivity in

- A) carbohydrates B) proteins
 C) DNA D) RNA

Solution : (Correct Answer: B)

Few bacteriophages when grown in bacterial culture containing ^{35}S . It was observed that radioactivity was incorporated into the cysteine and methionine amino acids of proteins and thus these amino acids with ^{35}S formed the proteins of phage.



141. Which of the following statement is correct about DNA polymerase?

- A) DNA polymerase can synthesize mRNA in the 3' to 5' direction.
 B) DNA polymerase can synthesize DNA in the 5' to 3' direction.
 C) DNA polymerase can synthesize mRNA in the 5' to 3' direction.

- D) *DNA* polymerase can synthesize *DNA* in the 3' to 5' direction.

Solution : (Correct Answer: B)

DNA polymerase is an enzyme which is involved in the replication and repair of *DNA*. It synthesizes new *DNA* strands using a *DNA* template in the 5' – 3' direction.

142. Lactose operon produces enzymes

- A) *b*–galactosidase, permease and glycogen synthetase.
 B) *b*–galactosidase, permease and transacetylase.
 C) permease, glycogen synthetase and transacetylase.
 D) *b*–galactosidase, permease and phosphoglucose isomerase.

Solution : (Correct Answer: B)

Lactose operon (model proposed by Jacob and Monod) produces three enzymes -*b*– galactosidase (*z*), permease (*y*), transacetylase (*a*).*b*- λ ; Galactosidase (*z*) is responsible for the splitting of lactose into glucose and galactose. Permease (*y*) is required in entry of the lactose/galactose. Transacetylase (*a*) transfers an acetyl *CoA* to *b*– Galactosides.

143. Choose the correct statement.

- A) Transcription and translation occur in same compartment for prokaryotes
 B) Monocistronic RNA - more than one structural genes under single promoter
 C) Introns and exons both code for protein synthesis
 D) In prokaryotes, splicing and tailing occurs before translation.

Solution : (Correct Answer: A)

Prokaryote are microscopic unicellular creatures. They possess simplest type of organization. In these cells, the process of transcription and translation occurs in the same compartment.

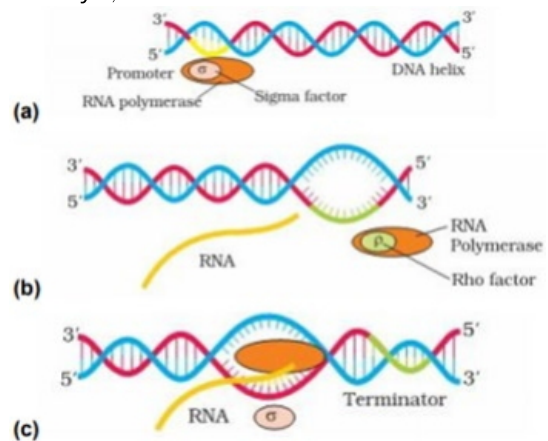
144. Codons of glycine are

- A) *CCU, CCC, CCA, CCG*
 B) *CGU, CGC, CGA, CGG*
 C) *GGU, GGC, GGA, GGG*
 D) *ACU, ACC, ACA, ACG*

Solution : (Correct Answer: C)

The codons of glycine are *GGU, GGC, GGA, GGG*. A genetic code is the smallest unit to code one amino acid. This genetic code is defined as nucleotide sequence of nitrogenous bases, which specifies the amino acid sequence in a polypeptide molecule.

145. Identify *a, b* and *c*



- A) (a) Elongation, (b) Termination, (c) Initiation
 B) (a) Initiation, (b) Termination, (c) Elongation
 C) (a) Initiation, (b) Elongation, (c) Termination
 D) (a) Termination, (b) Elongation, (c) Initiation

Solution : (Correct Answer: B)

The process of transcription occurs in three stages- initiation, elongation, followed by termination. The process occurs in the 5' – 3' direction. During initiation the complex of *RNA* polymerase, sigma factor and promoter is present at the beginning. It reaches in the middle during elongation. Finally, the complex is released at the time of termination.

146. In Meselson and Stahl's experiments, heavy *DNA* was distinguished from normal *DNA* by centrifugation in

- A) *CsOH* gradient B) $14\text{NH}_4\text{Cl}$
 C) $15\text{NH}_4\text{Cl}$ D) *CsCl* gradient

Solution : (Correct Answer: D)

In Meselson and Stahl's experiments, heavy *DNA* was distinguished from normal *DNA* by centrifugation in *CsCl* gradient. When *DNA* is mixed with caesium chloride it will settle down at a particular height in centrifugation and heavier one higher up.

147. Allelic sequence variations where more than one variant (allele) at a locus in a human population with a frequency greater than 0.01 is referred to as

- A) Incomplete dominance B) Multiple allelism
 C) *SNP* D) *DNA* polymorphism

Solution : (Correct Answer: D)

Allelic sequence variation has traditionally been described as a *DNA* polymorphism if more than one variant (allele) at a locus occurs in human population with a frequency greater than 0.01. In simple terms, if an inheritable mutation is observed in a population at high frequency, it is referred to as *DNA* polymorphism.

148. Who proved that *DNA* is basic genetic material?

- A) Mendel B) Watson
 C) Boveri and Sutton D) Hershey and Chase

Solution : (Correct Answer: D)

Hershey & Chase (1962) discovered that *DNA* is the genetic material of *E. coli*. Some virus made to grow on culture containing radioactive sulphur in

ome on radioactive phosphorus. Findings in aated genetic material.

149. Under which of the following conditions will there be no change in the reading frame of following *mRNA* 5' AACAGCGGUGCUAUU3'

- A) Insertion of G at 5th position
- B) Deletion of G from 5th position
- C) Insertion of A at G at 4th and 5th positions respectively
- D) Deletion of GGU from 7th, 8th and 9th positions

Solution : (Correct Answer: D)

150. Assertion : " DNA finger printing" has become a powerful tool to establish paternity and identity of criminals in rape and assault cases.

Reason : Trace evidences such as hairs, saliva and dried semen are adequate for DNA analysis.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

Solution : (Correct Answer: A)

DNA finger printing has become a powerful tool due to its ability in tracing evidences of crime and to establish the paternity. These evidences can be collected from hairs, semen and saliva.

151. Which of the following statements is correct?

- A) Adenine does not pair with thymine
- B) Adenine pairs with thymine through two H—bonds
- C) Adenine pairs with thymine through one H—bond
- D) Adenine pairs with thymine through three H—bonds

Solution : (Correct Answer: B)

Adenine pairs with thymine through two H—bonds

152. AGGTATCGCAT is a sequence from the coding strand of a gene. What will be the corresponding sequence of the transcribed *mRNA*?

- A) AGGUAUCGCAU
- B) UCCAUAGCGUA
- C) ACCUAUGCGAU
- D) UGGTUTCGCAT

Solution : (Correct Answer: A)

153. Which one of the following pairs of nitrogenous bases of nucleic acids, is wrongly matched with the category mentioned against it ?

- A) Guanine, Adenine - Purines
- B) Adenine, Thymine - Purines
- C) Thymine, Uracil - Pyrimidines
- D) Uracil, Cytosine - Pyrimidines

Solution : (Correct Answer: B)

(b) : The two DNA chains are held together by hydrogen bonds between their nitrogenous bases. Adenine (A), a purine of one chain lies exactly opposite thymine (T), a pyrimidine of the other chain. Similarly, cytosine (C), a pyrimidine lies

opposite guanine (G), a purine. Three hydrogen bonds occur between cytosine and guanine ($C \cong G$) at positions 1', 2', and 6' and two hydrogen bonds between adenine and thymine ($A = T$) at positions 1' and 6'.

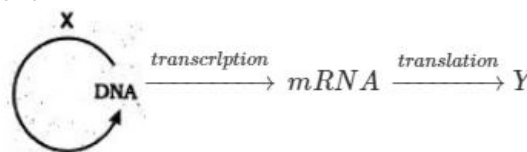
154. Identify the correct order of organisation of genetic material from largest to smallest.

- A) Genome, chromosome, gene, nucleotide
- B) Chromosome, genome, nucleotide, gene
- C) Chromosome, gene, genome, nucleotide
- D) Genome, chromosome, nucleotide, gene

Solution : (Correct Answer: A)

(a) : In genome all the genes are contained in a single set of chromosomes. The instructions in our genome are present in the form of DNA. DNA has a complicated structure in the form of a double helix. Single strands of DNA are coiled up into structures called chromosomes. Within the chromosomes, segments of DNA are "read" together to form genes. Thus, a gene is a segment of DNA or chromosome situated at a specific locus (gene locus) which carries coded information associated with a specific function and can undergo crossing over as well as mutation. A nucleotide is the basic unit of DNA made up of a pentose sugar, phosphoric acid and a nitrogenous base.

155. By using diagram, select the correct option for X and Y



- A) Replication Protein
- B) Cell division Protein
- C) Replication Amino acid
- D) t-RNA r-RNA

Solution : (Correct Answer: A)

156. The association of histone H_1 with a nucleosome indicates that

- A) DNA replication is occurring
- B) the DNA is condensed into a chromatin fibre
- C) the DNA double helix is exposed
- D) transcription is occurring.

Solution : (Correct Answer: B)

(b) : Histones help in packaging of DNA. In eukaryotes, DNA packaging is carried out with the help of positively charged basic proteins called histones. Histones are of five types - H_1 , H_2A , H_2B , H_3 and H_4 . H_1 is attached over the linker DNA. Histone contains a large proportion of the positively charged (basic) amino acids, lysine and arginine in their structure. DNA is negatively charged due to the phosphate groups on its backbone. The result of these opposite charges is strong attraction and therefore, high binding affinity between histones and DNA.

157. Select the correct option. Direction of reading of RNA synthesis the template DNA strand

Direction of <i>RNA</i> synthesis	Direction of reading of the template <i>DNA</i>
(a) 5' – 3'	3' – 5'
(b) 3' – 5'	5' – 3'
(c) 5' – 3'	5' – 3'
(d) 3' – 5'	3' – 5'

A) (a) B) (b) C) (c) D) (d)

Solution : (Correct Answer: A)

(a) : RNA polymerase initiates and extends the *RNA* (chain elongation) and functions always in 5' to 3' direction. The structural component of *DNA* has 3' to 5' polarity. It is also called template *DNA* strand or antisense (–) strand.

158. Which of following is not salient feature of human genome ?

- A) Human genome contains 3164.7 million nucleotide base.
- B) The function are unknown for over 50 percent of the discovered genes.
- C) More than 2 percent of the genome codes for proteins.
- D) Chromosomes 1 has most genes and they has the fewest.

Solution : (Correct Answer: C)

159. Formation of *RNA* strand on template *DNA* is

- A) Replication .
- B) Transcription
- C) Protein synthesis
- D) Synthesis of primer

Solution : (Correct Answer: B)

160. *RNA* as genetic material in

- A) *TMV* virus
- B) Cyano bacteria
- C) Clado phora
- D) Sieve cells

Solution : (Correct Answer: A)

161. The modern concept of gene is

- A) A segment of *DNA* capable of crossing over
- B) A functional unit of *DNA*
- C) A segment of *DNA*
- D) A segment of chromosome

Solution : (Correct Answer: B)

(b) A gene is a segment of *DNA* which contains the information for one enzyme or one polypeptide chain coded in the language of nitrogenous bases or the nucleotides.

162. Adenine is 30% then what would be the % of guanine

- A) 10%
- B) 20%
- C) 30%
- D) 40%

Solution : (Correct Answer: B)

(b) The percentage of *A + G* equals 50% so percentage of guanine is 20%.

163. *DNA* is double helix and

- A) Complementary and parallel
- B) Complementary and antiparallel
- C) Without super coils
- D) Always circular

Solution : (Correct Answer: B)

(b) The two strands (*polynucleotide chains*) of Double helix are antiparallel complementary.

The sequence of nitrogenous bases in one determines the sequence of nitrogenous bases in the other.

164. Transcription of *DNA* is aided by

- A) *RNA* polymerase
- B) *DNA* polymerase
- C) Exonuclease
- D) Recombinase

Solution : (Correct Answer: A)

(a) Enzyme involved in transcription is *RNA* polymerase -II which consist five polypeptide $\alpha, \beta, \beta', \omega$ and σ .

165. On which part of *tRNA*, amino acid gets linked

- A) Anticodon loop
- B) Variable loop
- C) 3' end
- D) 5' end

Solution : (Correct Answer: C)

(c) AA-Binding site- It lies at the 3' end. Amino acid binding site and anticodon are the two recognition sites of *tRNA*.

166. *DNA* strand is directly involved in the synthesis of all the following except

- A) *tRNA* molecule
- B) *mRNA* molecule
- C) Another *DNA* strand
- D) Protein synthesis

Solution : (Correct Answer: D)

(d) *DNA* does not move to the site of protein synthesis to directly guide the process. Instead, it transfers its information to *mRNA* molecules which move to the ribosomes to direct protein synthesis.

167. Repressor protein is formed from

- A) Repressor gene
- B) Structural gene
- C) Operator gene
- D) Regulatory gene

Solution : (Correct Answer: D)

(d) Regulator gene controls the activity of operator gene by producing repressor molecules or it codes for repressor protein (gene regulatory protein).

168. Property of a codon for always coding a specific amino acid is called

- A) Degenerate
- B) Non-ambiguous
- C) Non-overlapping
- D) None of the above

Solution : (Correct Answer: B)

(b) There is no ambiguity regarding genetic code, i.e., one codon will code for same specific amino acid in all organisms.

169. *DNA* is not directly involved with the synthesis of the following

- A) *m* – *RNA*
- B) *r* – *RNA*
- C) *t* – *RNA*
- D) Protein

Solution : (Correct Answer: D)

It's Obvious

170. Synthesis of any protein in a cell determines

- A) Types of ribosomes
- B) Mitochondria
- C) Sequence of nucleotides in *DNA*
- D) Sugar and phosphate of *DNA*

Solution : (Correct Answer: C)

It's Obvious

171. Which one of the following codons codes for the same information as *UGC*

A) *UGU* B) *UGA* C) *UAG* D) *UGG*

Solution : (Correct Answer: A)

(a) *UGC* and *UGU* both codons are responsible for cysteine amino acid.

172. The arrangement of three bases in the genetic code signifies a specific

A) Protein B) Amino acid
C) Plasmid D) Nucleic acid

Solution : (Correct Answer: B)

(b) Amino acid is coded by a group of three bases called as codon.

173. In Operon concept, the regulator gene regulates chemical reactions in the cell by

A) Inactivating enzymes in the reaction
B) Inhibiting transcription of *mRNA*
C) Inhibiting migration of *mRNA* into cytoplasm
D) Inhibiting the substrate in the reaction

Solution : (Correct Answer: A)

It's Obvious

174. The double helix model of Watson and Crick is known as

A) *C – DNA* B) *B – DNA*
C) *Z – DNA* D) *D – DNA*

Solution : (Correct Answer: B)

(b) *B – DNA* is the right handed coiled and normal type of *DNA*.

175. Eukaryotic *RNA* polymerase *III* catalyse the synthesis of

A) *m RNA* B) *t RNA*
C) *18.5 rRNA* D) Introns

Solution : (Correct Answer: B)

It's Obvious

176. Phosphorus is present in

A) Protein B) *DNA*
C) *RNA* D) Both *DNA* and *RNA*

Solution : (Correct Answer: D)

It's Obvious

177. Bacteria were grown in a medium containing heavy isotope of nitrogen (N^{15}) for many generations and all their *DNA* contained many heavy nitrogen only. A bacterium of this type was transferred to normal medium and allowed to duplicate. After two divisions of heavy *DNA* is likely to be that

A) Only one daughter cell will have heavy *DNA*
B) Two daughter cells have normal *DNA* and other two have both normal and heavy *DNA*
C) All daughter cells have heavy *DNA*
D) Half daughter cells have heavy *DNA* and other half have normal *DNA*

Solution : (Correct Answer: B)

(b) After two divisions, four bacterial cells are formed in which two daughter bacterial cells have

normal *DNA* and other two have both normal and

heavy *DNA* because *DNA* replicates by semiconservative method in bacteria.

178. The length of a helix of *DNA* is

A) $10\text{\AA}$ B) $20\text{\AA}$
C) $30\text{\AA}$ D) $34\text{\AA}$

Solution : (Correct Answer: D)

It's Obvious

179. The element absent in *RNA* is

A) Nitrogen B) Sulphur C) Oxygen D) Hydrogen

Solution : (Correct Answer: B)

It's Obvious

180. Isolation and purification of specific *DNA* segment from a living organism was achieved by

A) Crick B) Nirenberg
C) Khorana D) Beckwith and his colleagues

Solution : (Correct Answer: B)

It's Obvious